



Microgravity propellant gauging using modal analysis

Kassia Schraufnagel

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

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Current microgravity propellant gauging technology is inaccurate and not universally applicable for use in future spacecraft. A new method of gauging propellant in microgravity environments that uses modal analysis is discussed. This method, Modal Propellant Gauging (MPG), measures the structural acoustic resonant frequencies of the tank to determine the added mass of the propellant. This is achieved by comparing the input white noise frequencies with the output resonant frequencies through a Frequency Response Function (FRF). From the FRF, the modal parameters of the tank can be determined and consequently, the amount of liquid in the tank. This method is discussed in the context of fluid transfer conditions to determine the success of MPG in a dynamic setting.

I. INTRODUCTION

The ability to measure fuel is a vital part of rocket technology. Too much fuel can be expensive between the extra mass and price of propellant. Not enough fuel can be dangerous and leave the craft stranded. In order for humanity to have a permanent presence in space outside of low Earth orbit (LEO), refueling stations need to be developed, and these will depend on accurate propellant gauging [1].

Multiple types of fuel gauging techniques are currently being used. The most common technology, bookkeeping, involves subtracting the amount of expended fuel from the initial amount [2]. However, the lack of gravity causes the propellant to slosh around, bubble, and stick to the walls of the tank, leading to poor measurements [3]. While the amount of propellant in a near-full tank is measured with decent accuracy, the measurement error propagates every time fuel is expended, causing measurement error margins up to 10%.

Another common type uses a sensor that determines if it is immersed in the liquid or not. These “wet-dry” sensors can only determine if liquid is at the specific location of the sensor, but theoretically work with both storable and cryogenic propellants [4]. However, in low gravity conditions, there is potential for the liquid propellant to stick to the sensor through surface tension, leading to false indications of liquid presence.

Some other methods include measuring the gas-liquid surface. One type of these sensors are “level sensors,” which measure the location of the surface [4]. Others, “pressure sensors,” measure the difference in pressure at the top and bottom of the tank to calculate the depth of liquid, but can only be used in environments with sufficient gravity [4]. Another method involves “ultrasonic sensors,” which send ultrasonic pulses through the liquid and measure the echo [4]. While this type can be used in microgravity, the liquid configuration must be known, forcing a need for fluid containment devices when in mi-

crogravity conditions.

Capacitance probes can also be used. These probes are filled part of the way with liquid, depending on how much liquid is in the tank [4]. The amount of liquid in the probe changes the probe’s measured capacitance, allowing it to approximate the amount of liquid present. However, these probes are unable to perform well in microgravity conditions and need configuration control.

Many of these methods involve large margins of error, can only work with certain types of propellant, or are unable to be used under gravitational influence. The large margins of error force many spacecraft to decrease their lifetimes, leading to many satellites being retired with plenty of fuel left, an expensive sacrifice of potential revenue for many companies. The technologies that require hardware in the tank create the potential for leakage, a problem for cryogenic propellants. Additionally, the inability to work with multiple types of propellant or without gravity prevents many of these technologies from being more universally usable. A new method of measuring fuel in microgravity is needed.

An alternative method is Modal Propellant Gauging (MPG), which uses the structural acoustic resonant frequencies of the tank to determine the amount of propellant inside. This non-invasive technology has the potential to be more accurate and is specifically designed to operate in microgravity.

The Modal Propellant Gauging - Propellant Refueling and On-Orbit Transfer Operations (MPG-PROTO) experiment tests dynamic MPG technology. Specifically, the experiment is the first of its kind to test MPG on a tank that contains a changing mass of fluid. The purpose of the experiment is to determine the success of MPG during fluid transfer conditions. To do so, modal analysis is used to analyze the resonant frequencies of the tank according to the mass of fluid in the tank.

To explain how the amount of fluid in PROTO is determined, an overview of modal analysis and the theory behind MPG will first be provided. This information

will contextualize how data is collected and interpreted. Then, the PROTO experiment will be explained, along with the experimental design, testing environments, and experimental campaigns. Finally, MPG technology on PROTO is discussed, involving experimental and modeled data. This section will cover how the experiment's data predicts the fill fraction of fluid in the tank.

II. OVERVIEW OF CONCEPTS

Modal analysis involves measuring the frequency of vibrations in a structure to determine the modes at which the structure resonates. These modes are used to calculate modal parameters, such as frequency, damping, and mode shape [5]. Some of the more common applications of modal analysis include testing structural integrity, finding loose components, analyzing constraints and structural loading, and measuring modes and determining how to move them [6]. Even amongst this wide range of applications, the techniques used in modal analysis are similar for all of them.

To understand how modal analysis works in the PROTO experiment, a brief introduction to modal analysis is covered, then a more specific summary of its application in MPG is discussed.

A. Modal Analysis Basics

To understand modal analysis, first consider a simple mass-spring system. The system is excited by displacing and releasing the mass. In this scenario, the system oscillates due to being underdamped, meaning the damping factor is relatively low. The motion of the whole system can be represented by eq. (1):

$$m\ddot{x} + c\dot{x} + kx = f(t), \quad (1)$$

where m is the mass, x is the mass displacement from equilibrium, c is the damping coefficient, k is the spring constant, and $f(t)$ is a time-dependent function describing the system forcing function [5]. The mass oscillates at the same frequency as the forcing function.

All systems have a natural frequency ω , which is a function of mass and the spring constant:

$$\omega = \sqrt{\frac{m}{k}}. \quad (2)$$

The roots of eq. (1) are complex, where the imaginary root defines the damped natural frequency, which is commonly also referred to as the oscillatory rate. The real root of the equation gives the decay rate, which is the rate at which the amplitude of each oscillation slowly dies away due to the damping in the system.

The forcing function excites the system, creating oscillations at a designated frequency. Using the equation of

motion, the frequency response of the system, also called the transfer function, is determined. This describes the frequency at which the mass oscillates due to the forcing function or excitation. When the forcing frequency matches the natural frequency of the system, the forcing frequency is called the resonant frequency. At the resonant frequency, the amplitude is a maximum and is only constrained by the damping of the system [5].

In any vibrating physical system, there are modes. These modes refer to the shape deformation of the structure and are distinct for every frequency and damping coefficient [5]. Consider a wave in a box. For the first mode of the standing wave, there is only half a period oscillating between peak and trough. The second mode has a full period at half the original wavelength, the third mode has one and a half periods at a third of the original wavelength, and so on. These kinds of modes occur in all physical systems and have different mode shapes.

Since a physical system is typically observed as the oscillation amplitude over time, a Fast Fourier Transform (FFT) is used to convert this data into the oscillation amplitude over frequency [7].

Noether's Theorem states that there is a conservation law for every case of continuous symmetry in a physical system [8]. In this case, 'symmetry' refers to a transformation of the system that leaves the system physically unchanged. Since, by switching between the time and frequency domains through the FFT, the system is not physically changed, Noether's Theorem applies and conserves time and energy. Parseval's Theorem then verifies that the signal energy is conserved through the FFT, allowing this conversion to take place without data loss [9].

A common way to analyze the frequency response of a system is to use a Frequency Response Function (FRF), which is the ratio of the FFT of the output frequency to the FFT of the excitation frequency [5]. The transfer function is found by taking an FRF of the data, which divides out noise common between the input and output frequencies, leaving just the modes at which the structure resonates.

B. Modal Analysis Techniques

In a physical system, there are multiple ways to introduce and measure vibrations in a system. To vibrate the system, an excitation function is needed. Some of these mechanisms can include impactors, like a hammer, or shakers, like a shake table [5]. Impactors result in short, pulsed vibrations, but cover a wide range of frequencies. Shakers, however, provide an excitation over a longer period of time [5].

To measure the output frequency that the physical structure operates at, a sensor is needed. A common type of sensor is piezoelectric transducers, which generate electrical output when they vibrate [5].

Many of these systems are analyzed in terms of input

and output frequencies to determine the physical characteristics that create the changes between the two. Consequently, computer models of these physical systems are often created. These models are a mesh form of the physical structure, where each node is able to vibrate according to the frequencies of the structure [7]. These Finite Element Models (FEM) are complicated models based on many of the structural and physical characteristics of the system being modeled. These computer models are often compared with experimental data. Then, based on the differences between the results, either the experimental setup is updated to optimize the data or the FEM is updated to more accurately represent the physical structure and match the experimental data. Through this process, FEMs are developed to be accurate in representing the vibrations in the physical structure.

C. Theory of Modal Propellant Gauging

In MPG, the physical structure being analyzed is a tank containing a volume of liquid. For any system, however, the vibrations of a physical structure are heavily dependent on the characteristics of that structure. Some of the more important of these characteristics include the structure's surface stiffness and mass [5]. Both of these characteristics can easily change depending on the precise tank used, but neither is affected by the amount of liquid present in the tank. However, in microgravity, the liquid in the tank sticks to the tank walls due to surface tension [10]. This is particularly relevant in a spherical tank, such as the experiment tank used in PROTO. Consequently, whatever liquid is in the tank effectively adds to the mass of the tank. A larger tank mass has more damping, leading to lower resonant frequencies. By measuring the resonant frequencies of the tank, the added mass of the tank can be determined, and from here, the fill fraction of the tank.

To determine the resonant frequency of the tank, the input vibrations and the output vibrations need to be compared. Since the measurements of the two sensors are in terms of amplitude and time measured, an FFT is performed on the sensor output to obtain the data in terms of amplitude and frequency. Then, an FRF is performed to divide the input noise out of the data. Consequently, the FRF gives information on only the amplitude and the frequencies of the vibrations of the tank. From this information, the tank's modal parameters can be determined [11], allowing for derivation of the mass of the tank. The tank mass is then used to determine the fill fraction of the tank.

MPG uses modal analysis to determine the tank fill fraction, which is the amount of liquid or fuel in a tank. In order to do this, a lead zirconium titanate (PZT) patch, a type of piezo-electric actuator, is attached to the tank [12]. This actuator functions as a shaker input source to create the vibrations. These vibrations are generated by a white noise generator so that frequencies

ranging from 20 Hz to 20 kHz are input into the tank. Then, the electrical signals from the white noise generator are converted into mechanical compressions in the actuator [12]. These compressions spread throughout the tank, causing it to vibrate. Since the white noise input causes such a wide range of frequencies, the tank vibrates at its own resonant frequency as long as it is present in the range of frequencies provided.

Another PZT patch is placed next to the actuator, this time acting as a sensor. This sensor is compressed and elongated due to the vibrations in the tank caused by the actuator since they are directly next to each other. This way, the exact input frequencies of the actuator are measured and known. A third PZT patch is placed on the tank farther away. This sensor is compressed due only to the vibrations of the tank. When the sensor patches are compressed, electrical signals are measured, allowing the compressions to measure the vibrational frequencies of the tank [12].

III. EXPERIMENTAL OVERVIEW

The PROTO experiment tests MPG technology in a dynamic setting. This involves transferring fluids so that the fill fraction of the tank is not constant during testing to see if MPG is able to accurately predict each fill fraction. To better understand how the experiment works, a basic overview will be covered. This discussion will include a structural description of the experiment and an explanation of how the sensors relating to modal data are used. Then is a brief explanation of how the experiment is tested, and lastly, a description of how experiment campaigns are designed.

A. PROTO: Experimental Design and Overview

The goal of PROTO is to demonstrate the usability and determine the relative accuracy of MPG technology in a dynamic environment. A diagram of the experiment is shown in fig. 1.

The experiment is designed to achieve double containment, meaning that all liquid must be separated from the outside of the experiment and electronics by two waterproof seals. The first seal consists of the tanks and tubing. The second is created by the walls of the experiment.

The flow loop part of the experiment contains two tanks: an experiment tank and a catch tank. The spherical experiment tank is where MPG is tested and modeled on the spherical design of NASA's Lunar Gateway tank of the Artemis program [10]. The catch tank is the tank with which the liquid is transferred.

To operate, two fluid pumps are used on the experiment. The pumps allow fluid to be transferred between the two tanks. Both tanks are also connected at the

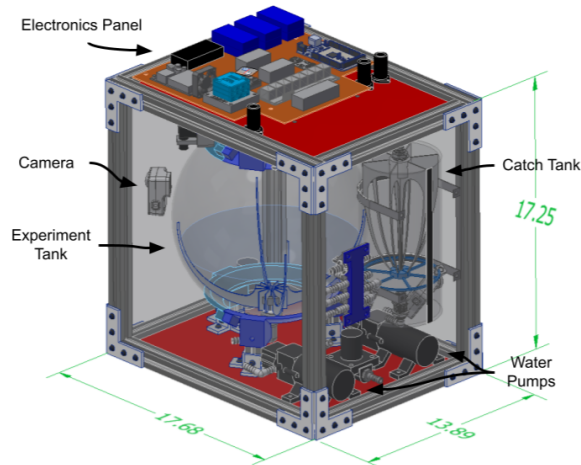


FIG. 1. A sketch of the PROTO experiment showing some of the major components.

opposite end through a vent port which allows them to equilibrate during tests to avoid building up pressure.

Outside of the flow loop, three cameras are used within the second layer of containment to monitor the experiment proceedings. These cameras can be programmed to power on and start recording based on flight timings to make sure all necessary data is recorded.

On top of the outside of the experiment is the electronics panel, which operates the entire experiment. The experiment can be programmed to be run in real-time through interactive operations or to run autonomously, making PROTO adaptable to different testing environments. The data acquisition for the experiment is controlled through SLICEWare.

The whole experiment was designed to be compatible with the two different companies through which the experiment is tested in microgravity: Zero-G and Blue Origin.

B. Testing Environments

To test the PROTO experiment in microgravity, it is operated through the aid of two companies: Zero-G and Blue Origin.

Zero-G uses parabolic flight campaigns to mimic microgravity environments [13]. On these flights, the plane flies upward at an angle near 45 degrees, then pulls out of the parabolic arc to mimic microgravity environments [13]. After about 20 to 30 s, the plane ends the maneuver and repeats the process. These flights allow the experiment to be tested in a microgravity environment while under the supervision of the experiment operators.

Each parabola presents at most 30 s of time in micro-

gravity, allowing for quick tests of the experiment. However, the plethora of parabolas permits many tests to be performed.

Since the experiment can be accompanied by people to operate it, it is allowed to be controlled in real-time, allowing for greater experimental flexibility and making the testing environment better for testing the experiment instead of gathering data.

The other vendor, Blue Origin, uses a New Shepard suborbital rocket to launch the experiment and other payloads into space in the crew capsule [14]. The crew capsule then lands, returning the experiment. The whole flight lasts about 15 min and gives the experiment about 5 min of zero-gravity time.

This method of testing allows for the liquid to be in a steadier state, rather than sloshing as much as in a Zero-G campaign. However, the 5 min of zero-gravity time on the rocket corresponds to about 3 min of usable time due to the sloshing caused by the separation of the crew capsule from the booster.

In this environment, the experiment has to be completely pre-programmed to operate before the flight. This means determining the specific campaign to perform in the 3 min of zero-gravity and a lot of flight simulations to make sure the experiment will operate as planned.

The PROTO experiment is designed to fit inside a Blue Origin double payload locker. The experiment is also adaptable to the other structural requirements of Blue Origin and Zero-G, allowing it to easily operate in both types of campaigns.

C. Designing Experiment Campaigns

In Blue Origin flights, the entire experiment campaign has to be designed in advance. With limited time in zero-gravity, the optimal way to collect data must be determined, planned, and coded into the experiment. This requires writing a concept of operations for the experiment.

To write a concept of operations, multiple factors must be taken into account. Since PROTO is supposed to test MPG in a dynamic environment, at some point in the 3 min of usable zero-gravity time, the experiment needs to transfer liquid between tanks. However, this time can be divided up into a fill, a drain, or both. Furthermore, as a control in the experiment, static data also needs to be taken, which is when the fill fraction does not change.

In addition to these decisions, the flow rates can make a difference. Specifically, a fast fill or drain rate could test the ability of MPG at a greater range of fill fractions and its ability to keep up with a fast pace. However, a fast rate could introduce more bubbles and turbulence, potentially decreasing the accuracy of the data.

Once flow rates are determined, the fill fraction of the tank at each moment of the flight needs to be calculated. This means determining the starting and ending amounts

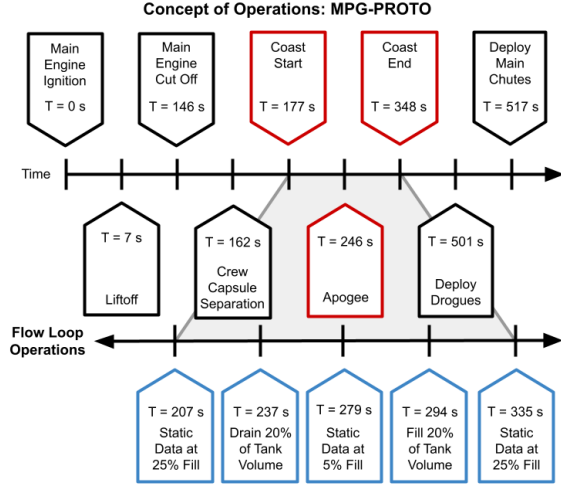


FIG. 2. A concept of operations used by PROTO on the P11 Blue Origin New Shepard mission [15]. The rocket events are shown on the top timeline in black and red, and flow loop events are on the bottom in blue. The red corresponds to all events occurring under zero-gravity.

of liquid in the experiment tank and the fill fraction at each time interval in between.

Having this information allows for better control of understanding how the experiment is coded and expected to operate. Furthermore, it allows for easier comparison to data attained in the actual campaign.

An example concept of operations used by PROTO is shown in fig. 2 [15]. This concept of operations was used by PROTO on the P11 Blue Origin New Shepard launch. During this campaign, data was gathered on MPG during a drain of the experiment tank, a fill, and when the tank level was static. This allowed for an initial insight into PROTO's accuracy in measuring fill fractions during each of these phases, getting initial data and allowing for planning future flights.

IV. MODAL PROPELLANT GAUGING

In many applications of modal analysis, experimental data is compared to data generated by a Finite Element Model (FEM), a computational version of the physical system. By comparing the two data sets, either the experiment setup or the FEM structure can be adjusted so that the two match. Once this is achieved, the FEM can be trusted to provide modal data without the need to rely on experimental tests.

Similar to many other modal experiments, the PROTO experiment utilizes both a FEM model and experimental data. The FEM of PROTO and its corresponding data are discussed next. Then, the experimental data is analyzed and compared to the FEM data.

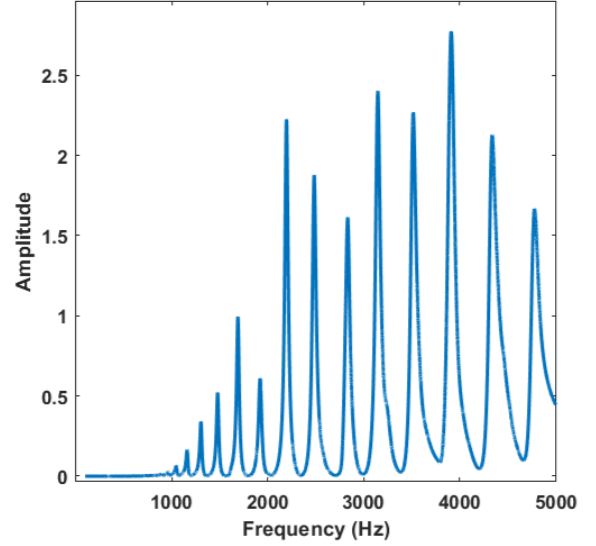


FIG. 3. The FEM of PROTO at a 25% fill fraction. The resonant frequency modes occur when the amplitude of the FRF peaks.

A. Finite Element Model of PROTO

To understand the data from PROTO, a FEM was created to predict the modal behavior of the experiment. This model has been tested at many different fill fractions. The fill fractions that will be focused on range from 0% to 60% in increments of 5%.

When an excitation function is used on the FEM, the entire mesh deforms according to the vibrational modes the physical structure would experience [5]. The nodes, where the mesh lines cross, measure these deformations and calculate the frequencies at which the structure vibrates. The FRF of the FEM data at a 25% fill fraction is shown in fig. 3 and is expected to resemble the experimental data at the same fill level.

Since these FRFs from the FEM can be calculated for many different fill fractions, the change in mode frequency can be tracked between fill fractions. To see how the added mass of liquid affects each mode, the frequency of the lowest mode is found for all of the tested fill fractions. This is accomplished by finding the peak in the data with the lowest frequency. Then, all of the frequencies of the lowest modes are plotted over the fill fraction of the tank in fig. 4.

The liquid in the tank can be approximated as an additional mass added to the tank, allowing the system to be effectively modeled the same but with a greater mass [10]. Consequently, this added mass greatly affects the frequencies at which the system resonates. In this case, the added mass from higher fill fractions causes the resonant frequencies of the tank to decrease.

The relationship between the frequency of the lowest mode and the fill fraction is shown to be predictable in

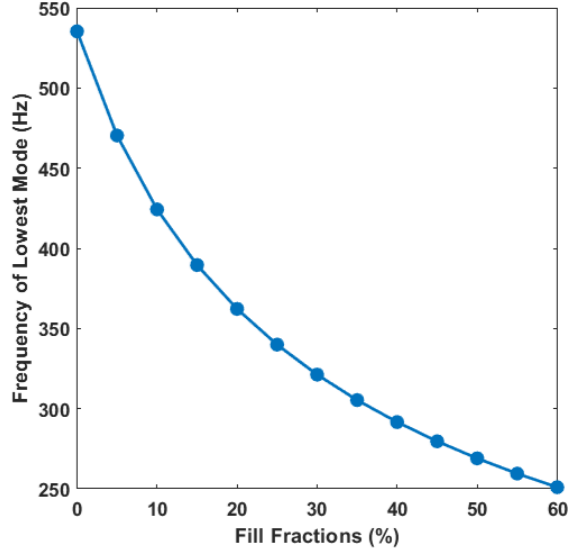


FIG. 4. The frequency of the lowest mode is found for each fill fraction and plotted over the respective fill fraction. As added mass increases due to larger fill fractions, the frequency of the lowest mode decreases.

fig. 4. MPG exploits the straightforwardness of this relationship by using it to predict the tank fill fraction from the resonant tank frequencies. While only the change in frequency of the lowest mode is shown, this same process can be completed for every mode of the tank, allowing highly accurate fill level approximations.

B. Analyzing data from PROTO

The data that will be analyzed was taken while the experiment was aboard a Blue Origin New Shepard rocket on the P11 mission. This flight used the experiment concept of operations shown in fig. 2 [1].

Data from a time slice during the first segment of the concept of operations will be analyzed. During this time slice, the tank has a fill fraction of about 25%. The FRF of the data is shown in fig. 5.

In fig. 5, the modal data peaks are present. However, the plot is considerably noisier than the FEM plot in fig. 3. While some of this difference is likely due to the noise introduced by the rocket, the FRF is used to divide out all noise. However, this only works under the assumption that the noise signals are equal in frequency and amplitude for both sensors. This is likely the case for the rocket but not for the pumps operating the experiment. By analyzing data in near time slices from when the pumps are not operating and when they are, little change is expected, where all peaks are anticipated to shift slightly downward in frequency. However, when the pumps turn on, more peaks are introduced and most of the data increases in magnitude. This is likely due to the

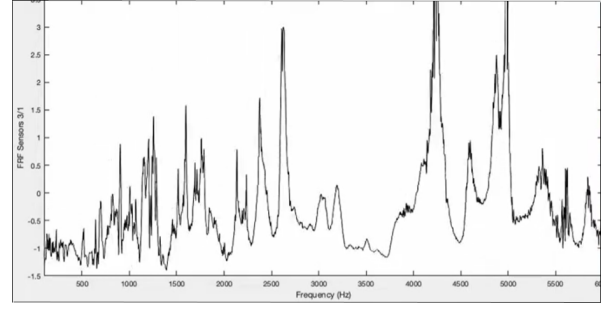


FIG. 5. Experimental data from PROTO while at a fill fraction of about 25%, taken during the P11 Blue Origin New Shepard flight campaign.

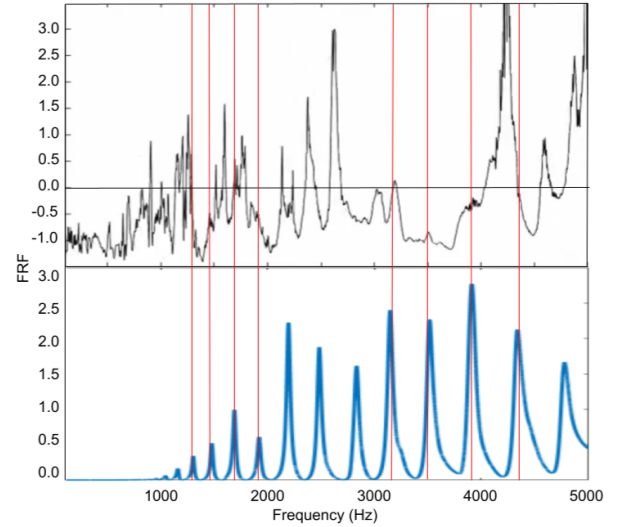


FIG. 6. A comparison of figs. 3 and 5. The experimental FRF from PROTO is on top and the FEM FRF is on the bottom. Red lines are used to show modal peaks from the FEM that appear clearly in the experimental data. The other peaks predicted by the FEM are present but obscured by noise in the experimental data.

magnitude of the noise not being equal at both sensors on the tank. Since one sensor is physically closer to the pumps, it receives a larger amplitude of noise from them, causing the noise not to be completely divided out during the FRF.

Due to this problem, not all of the peaks on the spectrum are modes of the tank's resonant frequencies. Consequently, determining the modes from the spectrum is best completed with an analytical model of the expected data. The model is derived from eq. (3):

$$\Omega = \frac{\omega_l}{\omega_0} = \sqrt{1 + Ar}, \quad (3)$$

where ω_l is the frequency of a mode at a fill fraction, ω_0 is the frequency of the same mode in the empty tank, Ω is the reduced frequency, r is the fill fraction of the

tank, and A is a constant based on the tank geometry and liquid distribution inside the tank. The reduced frequency Ω is a descriptor of the mode shift due to the fill fraction. The analytical model is fit with the FEM data to determine A . The accuracy of the fit is quantified by R^2 . Then, the analytical model can be applied to the experimental data.

Developing and fitting the analytical model is currently in progress, so instead, a different approach is used to explain the concept of mode fitting.

For an approximate fit of the FEM and experimental data, the two FRFs are compared in fig. 6. These two plots are quite different, hence the need for the analytical model.

In fig. 6, modes that visually appear in the FRF from both the experimental and the FEM data are marked with red lines. While the two plots shown in fig. 6 are very different, some of these modes appear in the experimental FRF. Since the modes appear in the experimental data, MPG is measuring the resonant frequencies of the tank. All of the extra peaks are noise, as predicted by the FEM. Since the peaks in the FRF of both the FEM and the experimental data are correlated, the FEM accurately predicted the modes measured in the experiment. Since the two data sets aligned, the FEM could be used in other data sets to predict the fill fraction and would be expected to be accurate.

While visually inspecting the data is not a clear or very accurate method, a correlation can still be found.

V. CONCLUSION

A new method of gauging propellant in microgravity environments uses MPG to relate a tank's structural acoustic resonant frequencies to the amount of liquid in the tank. In MPG, modal analysis is used to measure the frequencies at which the tank vibrates to determine the mass of the tank and the liquid. From this added mass, the amount of liquid in the tank is determined, allowing for improved accuracy in gauging propellant in space.

On the PROTO experiment, MPG is tested for the first time in a dynamic setting. This involves transferring liquid between two tanks and determining if MPG can accurately determine the amount of liquid in the experiment tank at each time slice.

To determine the accuracy of MPG, a computational model of PROTO is constructed and tested. This creates a computer-generated set of data describing the frequencies at which the experiment tank of PROTO is expected to vibrate. From this data, an analytical model can be used to find a correlation between experimental data and computational data. This correlation is used to determine the amount of liquid inside the tank at each time slice. By comparing the experimental data of PROTO with the data from the computational model, a correlation is found between the modal peaks. Consequently,

MPG is able to successfully determine the amount of liquid in the PROTO tank.

Future work involves finishing the development of this analytical model. This will result in higher accuracy in correlating the experimental data with the computational data. Furthermore, this will provide a method for determining the tank fill fraction from the data.

Future directions for the PROTO experiment involve updating the experiment to be compatible with other types of liquid. Currently, only water has been tested. Future research is directed to determining if the accuracy of MPG is affected by the behavior of propellant instead of water.

Aside from PROTO, research on MPG technology is being conducted on other experiments. One of these experiments seeks to further lengthen the amount of time MPG is tested in a microgravity environment. MPG-International Space Station (MPG-ISS) is a design for an experiment housed in a payload locker for the ISS. By flying on the ISS, the liquid used in the experiment will have more time to reach a steady state before data is taken, potentially reducing error from slosh.

Another of the experiments, known as MPG-Fiber Optic Sensing System (MPG-FOSS), is working to determine if the PZT patches used on PROTO and similar experiments could be replaced by a specific type of fiber optic cable originally developed by NASA. This experiment determines if these cables are sensitive enough to detect vibrations on a tank, allowing for the use of modal analysis, similar to the PZT sensors on PROTO.

Since MPG is being tested in microgravity conditions to improve fuel gauging methods for rockets and satellites, work is also being done to analyze its potential use on aircraft. Through a collaboration with Airbus, MPG-Airbus is an experiment that is working to determine the effect of tank orientation and PZT patch placement on the accuracy of MPG.

Another direction of research involves a technology similar to MPG that is used in the Microgravity Ullage Detection (MUD) experiment. This experiment that uses vibration amplitudes to determine the amount of ullage in a tank. Given the measured amount of ullage and the known tank volume, this information can be used to calculate the size and location of ullage in the tank.

With the success of MPG, it is being examined for application in many other uses and facing potential updates. Given the difficulties that lie with many of the current technologies being used to measure propellant in microgravity, a new solution, such as MPG, is needed soon. This is particularly urgent in the face of the revenue lost from decommissioned satellites and propositions for longer-distance space travel. Refueling stations for sustained presence in space will need improved technology, a solution presented by MPG [1]. An accurate fuel-gauging method is essential in creating a practical refueling station.

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